



Course Description	
COURSE NUMBER and NAME	CLD 400 Virtualization & Storage
UNITS	3
LENGTH OF CLASS	8 Weeks
COURSE DESCRIPTION	This course provides students an overview of cloud computing concepts, functional technologies, and hands-on experience through projects utilizing public cloud infrastructures (Amazon Web Services (AWS) and Microsoft Azure). Cloud computing services are being adopted widely across a variety of organizations and in many domains. This course introduces the cloud computing domain and covers topics focusing on storage and virtualization cloud infrastructures.
REQUIRED TEXT	None. Materials will be provided during the course.
INSTRUCTIONAL METHOD	Online / On-Campus

Summary of Graded Work and Assessments

Graded work and assessments offer students the opportunity to show the degree of mastery for each CLO. The following table shows how assessments and CLOs align (link).

Assignments	Totals	Weight	CLOs
Engagement and Professionalism (Rubric): Live Class Activities	160	16%	1, 2, 3, 4, 5, 6, 7, 8
Week 1 Discussion	50	5%	1
Week 2 Assignment	100	10%	2
Week 3 Discussion	50	5%	3
Week 4 Assessment	120	12%	4
Week 5 Discussion	50	5%	5
Week 6 Assignment	100	10%	6
Week 7 Assessment	200	20%	7, 8
Week 8 Final Presentation	170	17%	8
Total Points/Percentage	1000 Points	100%	



Course Policies

For Westcliff's course policies, please see the [Course Policies](#) document.

Discussion Requirements

For all discussions, the primary response is due by Thursday at 11:59 p.m. Pacific Time. The primary response must be at least 200 words in length and fully address the topic, demonstrating critical thinking and understanding. Each student must then also post a minimum of two responses to other students in the discussion by Sunday night at 11:59 p.m. Pacific Time. Each peer response must be at least 50 words in length and substantively engage with the other student's original post, continuing the discussion in a professional manner. If at any time information or material is brought in from an outside source or website, it must be properly cited following APA 7th edition guidelines and a full reference must be provided.

Assignment Requirements

Each assignment deliverable is specifically defined in the assignment instructions, such as page length, citations and references, audio or video, presentations, tables, etc. For all written assignments, the required page length does not include the cover or references pages. Refer to the specific requirements as stated in each assignment, and reach out to your instructor for additional information as needed. All graded submissions are due by Sunday at 11:59 p.m. Pacific Time.

All written work must adhere to APA 7th edition academic formatting requirements including core components such as the cover page, page numbers, headings, citations, 1" margins, paragraph indentations, left alignment, double spacing throughout, and the final references using hanging indents.

Participation Requirements

Students are required to attend each live class session either in person or virtually as stipulated in the course policies. Participation in the live class session is determined by actively engaging, answering or asking questions, providing comments, interacting in group activities, etc., as required by the instructor. Students who are unable to attend the live in-class or virtual sessions must follow the VCS submission requirements as stated in the Course Policies document.

Writing Center

The Westcliff University Writing Center is dedicated to providing quality support to students and faculty. From assignment review, to in-class workshops, to dissertation support, to publication help, the Writing Center is committed to empowering individuals to use the written language to articulate and disseminate knowledge.



Course Learning Outcomes (CLOs)

Learning outcomes are statements that describe significant and essential scholarship that students have achieved and can reliably demonstrate at the end of the course. Learning outcomes identify what the learner will know and be able to do by the end of a course – the essential and enduring knowledge, abilities (skills), and attitudes (values, dispositions) that constitute the integrated learning needed for successful completion of this course. The learning outcomes for this course summarize what students can expect to learn, and how this course is tied directly to the educational outcomes of the degree.

Course Learning Outcomes (CLOs)	PLOs
1. Discuss the differences between virtualization and cloud computing, and identify when a business would use them.	1
2. Apply business use case needs to different cloud server and storage virtualization needs.	2, 3
3. Compare different vendor virtualization services and solutions, and select specific business applications and selection criteria.	4
4. Compare different cloud service providers, and select specific business selection criteria.	4
5. Describe storage virtualization and discuss business use scenarios.	2
6. Evaluate data center storage management solutions.	2, 3
7. Discuss cloud storage center security and the top trends in cloud security affecting businesses.	2, 5
8. Develop and present a business solution to migrate data or services to the cloud.	1



Detailed Course Outline

The following outline provides important assignment details for this course, unit by unit. Students are responsible for all of the assignments given. Please refer to the Detailed Description of Each Grading Criteria in the syllabus for specific information about each assignment.

Week 1

Assignments to complete this week:

- Reading:
 - [Virtualization vs. Cloud Computing: What's the Difference?](#)
- Video:
 - [Virtualization](#)

Week 1 Live Class Activity

Watch the Virtualization video in class. Break into small groups depending on class size and research how IaaS functions using virtualization technologies. Discuss your findings in the live class.

Week 1 Discussion ([Rubric](#))

Infrastructure as a Service (IaaS) is one of the main foundations on which cloud computing functions. How does it actually work? Select an IaaS provider (e.g., Google, Amazon, Apple, GoDaddy, etc.) and research how they deliver cloud infrastructure services to customers. How many data centers do they have and where? As a business wanting to set up a server using one of these services, where will the server actually be located? If there are issues and the server is not working, how does it get fixed if the company cannot physically touch it?



Week 2

Assignments to complete this week:

- Reading:
 - [What's the Difference Between Type 1 vs. Type 2 Hypervisor?](#)
 - [What is Server Virtualization: The Ultimate Guide](#)

Week 2 Live Class Activity

In small groups depending on class size, review the article from this week's reading, [What's the Difference Between Type 1 vs. Type 2 Hypervisor?](#). In each group, discuss a business scenario in which each type would be appropriate. Also, discuss what the challenges may be in using each type for your scenario. Discuss your findings with the live class.

Week 2 Assignment ([Rubric](#))

Review the following article in this week's reading: [What is Server Virtualization: The Ultimate Guide](#). Select 2 of the main sections from the article:

- Why is server virtualization important?
- How does server virtualization work?
- What are the benefits of server virtualization?
- What are the disadvantages of server virtualization?
- Use cases and applications
- What are the types of server virtualization?
- Migration and deployment best practices
- Server virtualization management
- Vendors and products
- What's the future of server virtualization?

In a 2-3 page APA-formatted paper, summarize what you have learned from the 2 article sections about server virtualization that you selected to read. Present a brief business use case scenario in which server virtualization would be appropriate and address the pros and cons of deployment.



Week 3

Assignments to complete this week:

- Reading:
 - [Hyper-V vs. KVM Comparison](#)
 - [The VMWare vs. KVM Hypervisors Comparison](#)
 - [Choosing between Hyper-V vs. VMware](#)

Week 3 Live Class Activity

The articles listed in this week's reading compare Linux and Windows hypervisors from several different angles. In small groups depending on class size, create your own comparison tables showing the various benefits and features of each. You are welcome to bring in another hypervisor solution as well, if you desire. Share and discuss your comparisons with the live class. Additionally, discuss why a business would select one over another.

Week 3 Discussion ([Rubric](#))

There continues to be a lot of debate regarding which operating system has better security: Windows or Linux. When it comes to server virtualization, research the security of each operating system and share your findings with the class.



Week 4

Assignments to complete this week:

- Reading:
 - [AWS vs. Azure vs. GCP: The Cloud Platform of your Choice?](#)

Week 4 Live Class Activity

In small groups depending on class size, use the article for this week's reading and discuss the comparisons. Do you agree with the comparison results? Do you have any preferences for one vendor over another? Do you see any other comparison areas that would benefit a company to look at? Discuss your findings with the live class.

Week 4 Assessment ([Rubric](#))

The article reading for this week provides several comparisons and tables showing features of Amazon Web Services, Microsoft Azure, and Google Cloud Platform. Deciding which to use for a business depends on many different factors, such as what the cloud platform will be used for, what the business is currently using, cost, etc. Your final assessment for this course will be to present a business use case and propose a cloud solution. The first half of this course has focused on cloud virtualization technologies and the different vendor solutions available. The second half of the course will focus on cloud storage virtualization technologies and applications.

For this week, you will develop your business use case selection and problem determination. In a 3-4 page APA-formatted paper, select and introduce a business or organization and a specific need they may have that could best be addressed through cloud virtualization or storage technologies. For example, the company may have very outdated servers, they do not want to invest a large amount of money into purchasing new server hardware or housing the servers in their own environment, and they are interested in what the potential costs and benefits might be in setting up servers in a cloud environment. Another example may be that a company has a lot of data storage in-house, but due to business expansion to other locations, they want the data to be more accessible so they are interested in potential cloud data storage solutions.

For your paper, after introducing the business situation and need, use the research that you have performed during the first half of this course and build your recommendation. This week's paper includes:

- Business background and situation
- Solution research and clear recommendation with your justification and selection criteria

You are welcome to include comparison tables, pro and con lists, etc., as appropriate. Be sure to cite your sources and provide full references.

Your final paper will build on this week's work and include the solution design and implementation plan.



Week 5

Assignments to complete this week:

- Reading:
 - [Storage Virtualization](#)
- Video:
 - [What is Cloud Storage](#)

Week 5 Live Class Activity

Review this week's video. After watching the video, break into small groups depending on class size and research different cloud storage vendors (e.g., Google, AWS, Apple, Egnyte, etc.). Create a comparison table addressing cost, storage amounts, security, etc. Share and discuss your findings with the live class.

Week 5 Discussion ([Rubric](#))

Review the article for this week's reading: [Storage Virtualization](#). Why would a company want or need to use cloud storage virtualization? Are there any concerns in storing business information in the cloud? Are there any particular industries where storing information in the cloud could pose regulatory or security challenges?



Week 6

Assignments to complete this week:

- Reading:
 - [Data Center Storage - Impact of Cloud and IoT](#)
 - [Break Down Types of Data Center Storage](#)
 - [Cloud Backup Services Comparison](#)

Week 6 Live Class Activity

Cloud data centers are enormous facilities filled with many rows of drive and server cabinets. These environments require immense amounts of power, cooling, and security. Additionally, all of these systems must be backed up, mirrored, and systems in place to keep them running in case of power failure. For this activity, break into small groups depending on class size and research how specific companies like Google, AWS, Microsoft, etc., back up their data centers. How do they manage the backups, separation of customer information within the backups, etc.? Share your findings with the live class.

Week 6 Assignment ([Rubric](#))

For this assignment, research different tools that cloud storage service providers use to manage the vast arrays of drives in their environments. Also, research the current performance and trends of the types of hard drives in use: magnetic vs. SSD. Summarize your research in a 2-3 page APA-formatted paper. Be sure to provide citations and references for your sources.



Week 7

Assignments to complete this week:

- Reading:
 - [6 Cloud Security Challenges you Should Know About](#)
 - [The Biggest Cloud Security Challenges](#)
 - [Top 5 Cloud Security Concerns and How to Avoid Them](#)

Week 7 Live Class Activity

Using the articles in this week's reading, break into small groups depending on class size and focus on the security issues relating to virtualization and storage. Discuss how businesses should address these security concerns and what challenges they may have to overcome. Discuss your findings with the live class.

Week 7 Assessment ([Rubric](#))

For this week, you will design and plan the implementation of your business use case selection. Adding to your Week 4 Assessment paper, write the detailed design plan for your cloud solution addressing the business use case. Also, write out the implementation plan for the business to move to the proposed solution. Answer questions, such as cost, timeline, data migration, access rights, security, etc. Include tables or charts as appropriate. Your full combined paper will be 8-10 APA-formatted pages in length. Be sure to cite your sources and provide full references.

You will present your plan during the Week 8 live class.



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Week 8

Assignments to complete this week:

Week 8 Final Presentation ([Rubric](#))

Students will present their full plan as written in their Week 7 Assessment. PowerPoint presentations will be between 10-12 slides, and presentations will last approximately 15 minutes, depending on class size and available time.